5.7 Photosynthesis

i Know the structure of chloroplasts, including: envelope, stroma, grana and lamellar structure.

Light-dependent stage:

ii Understand the role of the thylakoid membranes in the light-dependent stage of photosynthesis.

iii Understand that the processes of cyclic and non-cyclic photophosphorylation result in the production of reduced NADP, ATP and oxygen.

Light-independent stage:

iv Understand the role of the stroma in the light-independent stage of photosynthesis.

v Understand how carbon dioxide is fixed by combination with 5C ribulose bisphosphate (RuBP) to form glycerate 3-phosphate (GP) using the enzyme ribulose bisphosphate carboxylase (RUBISCO).

vi Understand how reduced NADP and ATP from the light-dependent stage are used:

- to synthesise glyceraldehyde phosphate (GALP) from GP
- to regenerate 5C ribulose bisphosphate in the Calvin cycle (details of intermediate compounds are not required).

vii Understand how GALP is used as a raw material in the production of monosaccharides, amino acids and other molecules.

viii Understand the factors that limit photosynthesis including carbon dioxide, light intensity and temperature.